

Discrete Mathematics:

I. Mathematical Logic

1. Propositional Logic

Components: A formal system, truth values, rigorous rules.

Elements: propositional variables, propositional logic operators, evaluations

↓ represents propositions ↓ Connectives between variables ↓ True / False (Boolean value)

A variables itself is an atomic proposition.

Negation: $\neg \phi$, $T \rightarrow F$, $F \rightarrow T$. not-

Conjunction: $\phi \wedge \psi$, $(T, T) \rightarrow T$, else $\rightarrow F$. - and -

Disjunction: $\phi \vee \psi$, $(F, F) \rightarrow F$, else $\rightarrow T$. - or -

Conditional: $\phi \rightarrow \psi$, $(T, F) \rightarrow F$, else $\rightarrow T$. - imply -

Biconditional: $\phi \leftrightarrow \psi$, $(T, T) \rightarrow T$, $(F, F) \rightarrow T$, else $\rightarrow F$. - iff -

Exclusive-Or: $\phi \oplus \psi$, $(T, F) \rightarrow T$, $(F, T) \rightarrow T$, else $\rightarrow F$. - or - but not both

Precedence: $()$, $[\]$, $\neg$, $\wedge$, $\vee$, $\oplus$, $\rightarrow$, $\leftrightarrow$

Remark: " $\rightarrow$ " commonly used in compound logic, " $\Rightarrow$ " commonly used in proof

True table is a tool for analysis.

P	Q	R	...	$f(\phi, \psi, \dots)$
T	T	T	...	...
T	T	F	...	...
...	...	...	...	...
F	F	F	...	...

Mathematics Computer workflow Logical Circuits (Logical operators also obtain the same correspondance

T 1 High voltage

F 0 Low voltage

$P \rightarrow \neg Q \rightarrow \neg P$ $\begin{matrix} P \\ Q \end{matrix} \rightarrow P \wedge Q$ $\begin{matrix} P \\ Q \end{matrix} \rightarrow P \vee Q$

Inverter Gate And Gate Or Gate

relationships: connectives $\rightarrow$ Gates (In Circuits).

$\rightarrow$ Some special kind of compound propositions:

1. Tautology: Compound proposition that is always True, no matter the boolean value.

2. Contradiction: Compound proposition that is always False,

3. Contingency: Compound prop., whose correctness depends on the variables.

$\rightarrow$ Logical equivalence: p, q are two compound prop.s, if $p \leftrightarrow q$ is a tautology, we say p and q are logically equivalent. denoted as $p \equiv q$.

$p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$ According to the truth table.

$p \rightarrow q \equiv (\neg p) \vee q$; $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

Laws: Absorption Laws: $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

Distributive Laws: $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

De' Morgan's Laws: $\neg (\bigwedge_{i=1}^k P_i) \equiv \bigvee_{i=1}^k (\neg P_i)$, $\neg (\bigvee_{i=1}^k P_i) \equiv \bigwedge_{i=1}^k (\neg P_i)$

Congruence Properties: if $\phi \equiv \psi$, $\phi \circ \xi \equiv \psi \circ \xi$, $\circ \in \{ \wedge, \vee, \rightarrow, \oplus, \leftrightarrow \}$.

Satisfiability: A compound proposition is Satisfiable if there exists truth assignment(s) for propositional variable th makes it true.

Else: unsatisfiable. A truth assignment is called a solution.

(Checking satisfiability is a NP-incomplete question). DPLL $\wedge$ CDCL Algorithm are ways to boost efficiency

NP-Completeness: Non-determinant Polynomial Complete

2. Predicate Logic. components: predicates, quantifiers, quantifiers with restricted domains, logical equivalence, nested quantifiers.

predicates, $\forall x P(x)$: a variable that represents an entry in some domain. "Domain somehow determine the correctness of the predicate logic".

quantifiers: $\forall x P(x)$: $\left\{ \begin{array}{l} \text{Universal quantifier } \forall: \text{For all} \\ \text{Existential quantifier } \exists: \text{There exists.} \end{array} \right.$ $\forall x P(x)$: $P(x)$ holds for all values x in the domain. It is false when $\exists x: (\neg P(x))$
 $\exists x P(x)$: $P(x)$ holds for at least one x in the domain. It is false when $\forall x (\neg P(x))$.

Remark: $\forall x_i P(x_1, \dots, x_i, \dots, x_n)$ and $\exists x_j P(x_1, \dots, x_j, \dots, x_n)$ are "Predicates" rather than a statement. But why???

Remark: $\forall x P(x)$, $x \in \emptyset$ is a tautology because there won't be counterexample forever

$\exists x P(x)$, $x \in \emptyset$ is a contradiction. Because $\forall x (\neg P(x))$ $x \in \emptyset$ is a tautology.

Free / Bound variables: A variable is bound only by some quantifier whose application range directly include this variable

$\exists x_i$ (or $\forall x_i$) $P(x_1, \dots, x_n)$ is a predication logic only if: All the variables are bounded

Eliminate the restrictions: $\forall (x: Q(x, y)) P(x, y) := \forall x [Q(x, y) \rightarrow P(x, y)]$, $\exists (x: Q(x, y)) P(x, y) := \exists x [Q(x, y) \wedge P(x, y)]$.

Logical Equivalence: We say that $\phi \equiv \psi$, no matter which concrete predicates and domains are given, the truth values are coincide. For all the possibilities.

Laws: De'morgan's Laws: $\neg (\forall x P(x)) \equiv \exists x (\neg P(x))$ $\neg (\exists x P(x)) \equiv \forall x (\neg P(x))$

$\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$ $\exists x (P(x) \wedge Q(x)) \equiv (\exists x P(x)) \wedge (\exists x Q(x))$

$\forall x (P(x) \vee Q(x)) \equiv (\forall x P(x)) \vee (\forall x Q(x))$ $\exists x (P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$.

Nested Quantifiers: Order make sense.

3. Normal Forms: 1) Disjunctive Normal Form "Disjunction of Conjunctions".

(For prop.) Theorem: All the compound proposition is logically equivalent to some compound proposition. Proof: truth table!

2) Conjunctive Normal Form "Conjunction of Disjunctions".

Theorem: 1° Construct the truth table of $\neg \phi$, and its disjunctive normal form. then use de'morgan's law, get $\neg (\neg \phi)$.

A collection of logical operators is functional complete if any compound prop. is equivalent to a compound propositions using only connectives on the set.

$\{ \neg, \wedge, \vee \}$, $\{ \neg, \wedge \}$, $\{ \neg, \vee \}$ are all functional complete

Assume S_B is a collection of connectives constructed from S_A . then if S_A incomplete $\rightarrow S_B$ incomplete; if S_B is complete, $\rightarrow S_A$ is complete

3' Normal Forms: Prenex Normal Form: A statement is in Prenex normal form if be like:

(For pred.) $\beta_1, x_1, \dots, \beta_k, x_k, P(x_1, \dots, x_k)$, $\beta_i \in \{ \forall, \exists \}$.

We can Always transform a predicate to prenex normal form. Just move it directly. IN ORDER!

4. Inference Rules: Arguments := (premises $\rightarrow$ conclusions) (We won't consider the situation when premises $\equiv F$).

Argument forms:

" $\frac{P}{p \rightarrow q}$ "
" $\therefore q$."

1. Modus Ponens: if p , if $p \rightarrow q$, then q . $(p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology.

2. Modus Tollens: if $\neg q$, if $p \rightarrow q$, then $\neg p$. $((\neg q) \wedge (p \rightarrow q)) \rightarrow (\neg p)$ is a tautology.

3. Hypothetical Syllogism: $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology.

4. Disjunctive Syllogism: $p \vee q$ and $\neg p$, then q . $((\neg p) \wedge (p \vee q)) \rightarrow p$ is a tautology.

5. Addition: p is true, the $p \vee q$ is true. 6. Simplification: $p \wedge q$ is true, p is true. 7. Resolution: $p \vee q$ and $(\neg p) \vee r$, the $q \vee r$.

4' IR for predicate logic: Just translate them

Remark: Gödel's incompleteness Theorem says we can't infer all the theorem from a collection of premises (Axioms)